Multiplication law

1

a y^5 **b** $8x^9$ **c** $12y^7$ **d** x^{y+z}

e $-6a^{12}$ **f** $-10a^5$ **g** b^{13} **h** a^{b+c}

i $30y^8$ **j** $24u^4$ **k** $20b^{13}$ **l** x^{a+b+c}

2

a 3 **b** $5x^{12}$ **c** -2 **d** $6x^5$

Division law

3

a a^4 **b** $\frac{x^3}{5}$ **c** $6m^{11}$ **d** b^4

e j^7 **f** x^3 **g** f^3 **h** s^9

i $2k^5$ **j** $8g^7$ **k** $3p^3$ **l** $5l^6$

4

a 4 **b** 5 **c** 11 **d** 4

Power of a power law

5

a

i Yes **ii** Yes **iii** Yes **iv** No

v No

b

i False **ii** True

c 6

6

a j^{18} **b** w^8 **c** t^{12} **d** $27a^{12}$

e $125h^{21}$ **f** x^5y^{10} **g** $\frac{16}{h^8}$ **h** $\frac{a^6}{b^{15}}$

i $\frac{27x^6}{y^{15}}$ **j** x^{36} **k** $-125x^{12}$ **l** $-\frac{8a^{12}}{27b^6}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Zero index

8

a 18 **b** 1 **c** 1 **d** 1

e 9 **f** 1 **g** $\frac{3}{t^2}$ **h** 8

9

a 0 **b** 8 **c** 9 **d** 0

Mixed laws

10

a $x^{24}y^{12}$ **b** y^8 **c** p^{17} **d** $8y^6$

e m^8 **f** p^2 **g** $16x^{12}$ **h** $3p^9$

i 0 **j** -10 **k** 22 **l** x^{60}

m $16u^{10}v^4$ **n** $81a^8b^{20}c^4$ **o** $1125y^{16}$ **p** x^3

q $16x^3$ **r** c^{86} **s** $\frac{3a^2}{b}$ **t** -9

u 3^{a+6} **v** 3^{43a-21}

11 2^{16p}

© ClipSAT: By Mr. Mohamed Abdallah +966597688647